

## **Early Life History of Cavala preta, *Decapterus macarellus*, by the Larval Net Sampling**

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### **Abstract**

Larval samplings were carried out in S. Vicente Straight between October 2002 to September 2003, using larval net with 45 cm mouth diameter and 330 $\mu$  mesh. Towing depth and water temperatures were recorded. Surface water temperature was measured. All specimens were fixed in 10% formaline in the field and transferred to 80% ethanol for larvae after sorting in the INDP laboratory. Surface water temperature decreased from October to February (26°C to 21°C), then increased to September (to 27°C). Larvae and eggs of Cavala preta, *Decapterus macarellus*, was successfully identified. In a total of 6131 larvae collected from October 2002 to Mar 2003, 10 specimens belonged to the Carangidae family representing 5 taxas; 6 of them found in January, 2 in October and 2 in November. However, none of *Decapterus macarellus* was found during this period. From July and August samples, a large number of larvae and eggs were collected. Number of larvae per 1000m<sup>3</sup> increased in February and March in stations 4-6 near S. Vicente coast, when surface water temperature decreased. Further identification has been carried out. Detail discussion will be made after all larvae are identified.

### **Introduction**

The Cape Verde archipelago is located between 14° 50' and 17° 20' N of latitude and 22° 4.' And 25° 30' W of longitude, and consists of ten main islands and eight islets located approximately 600 Km off the western coast of Africa with a total land area of 4,033 Km<sup>2</sup>. Area of EEZ in Cabo Verde is 734,265 Km<sup>2</sup>, and the continental shelf that is narrow and 10,150 Km<sup>2</sup>. The climate is of the dry tropical sahelian type, which is under influence of the north-east trade wind, with moderate temperatures and narrow temperature range due to the Atlantic environment. During the hot and wet season (July to November), the average of temperature is 27°C and during the cold season is around 20°C. Rainfall is generally low over the entire archipelago.

Marine resources are very important and fishery has a great contribution economically and socially. In the area of marine biodiversity, more then 600 species of marine fish have been counted including migratory species.

Cavala preta is one of the most important fishery resources in Cabo Verde. Several studies have been done for this species although those focus to juvenile to adult stage after recruitment into fishery. Live history of this species has not been completed yet, especially early life stage. This report presented preliminary results of larvae of Cavala preta, *Decapterus macarellus*, at S. Vicente Straight in Cabo Verde.

### Material and methods

Larvae samples were collected monthly between October 2002 and September 2003 in the S. Vicente Strait (Figure 1) at 6 stations. Positions of stations are shown in Table 1. Oblique tows were made for all stations. Horizontal tows were done from Sts. 4 to 6. The sampling has been done using larval net with 45 cm mouth diameter and 330 $\mu$  mesh, during daytime.

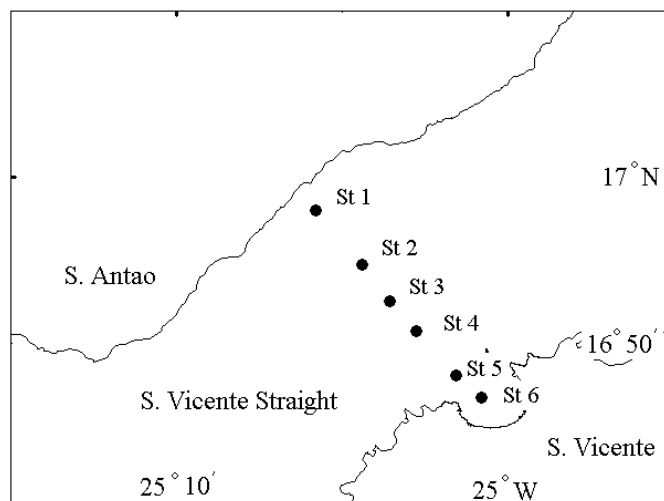


Figure 1. Stations of Larval sampling from October 2002 to September 2003.

Table 1. Position of larval sampling stations.

| Station No | Lat          | Long       | Depth(m) | Depth of Horizontal Tows |
|------------|--------------|------------|----------|--------------------------|
| 1          | 16° 59' 0 N  | 25° 05' 8W | 50m      |                          |
| 2          | 16° 57' 5 N  | 25° 04' 2W | 290m     |                          |
| 3          | 16° 56' 3 N  | 25° 03' 3W | 100m     |                          |
| 4          | 16° 55' 5 N  | 25° 02' 8W | 80m      | 60m, 40m, 20m, Surface   |
| 5          | 16° 54' 0 N  | 25° 01' 5W | 50m      | 40m, 20m, Surface        |
| 6          | 16° 53' 34 N | 25° 00' 8W | 20m      | 15m, surface             |

Towing depth and water temperatures were recorded by compact TD ATR-HR. Surface water temperature were also measured and transparency was measured using the Secchi disk.

All specimens were fixed in 10% formaline in the field and transferred to 80% ethanol after sorting in the INDP laboratory. After sorting, larvae identification was carried out, using various literatures listed in References.

### Results and Discussions

As seeing in changes of surface water temperature in Figure 2, temperature decreased from October to February (26°C to 21°C), then increased to September (to 27°C).

Larvae and eggs of Cavala preta, *Decapterus macarellus*, were shown in Figures 3 and 4. In a total of 6131 individuals fish from October 2002 to Mar 2003, 10 specimens belonged to the Carangidae family representing 5 taxas; 6 of them found in January, 2 in October and 2 in November. However, none of *Decapterus macarellus* was found during this period. From July and August samples, a large number of larvae and eggs were collected. Number of larvae per 1000m<sup>3</sup> was calculated in Table 2. Number of larvae increased in February and March in stations 4-6 near S. Vicente coast when surface water temperature decreased.

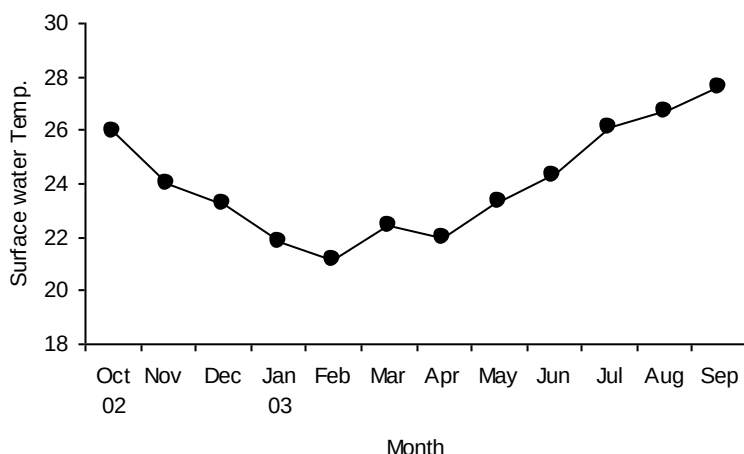


Figure 2. Monthly average surface water temperature.



Figure 3. Larvae of Cavala preta.

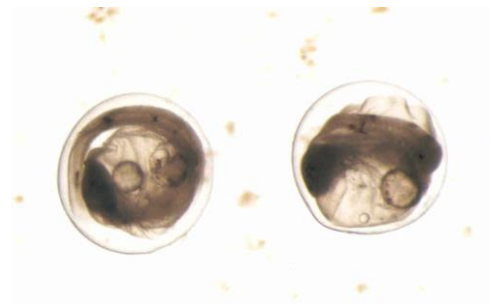


Figure 4. Eggs of Cavala preta.

Table 2. Number of larvae per 1000m<sup>3</sup>.

| Station | 2002  |       |        | 2003   |        |        |
|---------|-------|-------|--------|--------|--------|--------|
|         | Oct   | Nov   | Dec    | Jan    | Feb    | Mar    |
| 1       | 509   | 1,722 | 1,993  | 511    | 6,813  | 2,095  |
| 2       | 773   | 588   | 636    |        | 1,202  | 783    |
| 3       | 558   | 2,232 |        | 2,253  | 1,732  | 1,466  |
| 4       | 2,586 | 1,386 | 2,928  | 3,433  | 2,665  | 1,320  |
| 5       | 844   | 2,598 | 11,547 | 3,618  | 2,672  | 2,683  |
| 6       | 2,075 |       |        | 3,464  | 13,145 | 6,130  |
| Total   | 7,344 | 8,526 | 17,105 | 13,279 | 28,229 | 14,477 |

Results from horizontal samples were shown in Table 3. Table 3 showed a number of larvae per 1000m<sup>3</sup>.

Table 3. A number of larvae per 1000m<sup>3</sup> by the horizontal tows

| Month | St. | Depth(m) |      |      |      |
|-------|-----|----------|------|------|------|
|       |     | 0        | 20   | 40   | 60   |
| Oct   | 4   | 552      | 581  | 1566 | 1200 |
|       | 5   |          | 4072 | 2971 |      |
|       | 6   | 344      | 2121 |      |      |
| Nov   | 4   | 1833     | 1483 | 1775 | 701  |
|       | 5   | 2411     | 1954 | 328  |      |
|       | 6   | 1515     | 3343 |      |      |
| Dec   | 4   |          |      |      | 1490 |
|       | 5   | 3059     |      | 6197 |      |
|       | 6   | 1040     |      |      |      |
| Jan   | 4   | 1181     | 2176 | 3248 | 3152 |
|       | 5   | 1277     | 3773 | 5385 |      |
|       | 6   | 2922     | 3168 |      |      |
| Feb   | 4   | 1249     | 1627 | 977  | 2320 |
|       | 5   | 4906     | 4001 | 2123 |      |
|       | 6   | 1475     | 8520 |      |      |
| Mar   | 4   | 126      | 521  | 846  | 1950 |
|       | 5   | 1425     | 5406 | 1951 |      |
|       | 6   | 2043     | 1669 |      |      |

Unfortunately, at the moment identification work for rest of samples from April to September has been continued, thus complete discussions have to be waited.

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